

General Education Reflection

General education courses have enhanced my learning while developing as an engineer at Iowa State University. I have had the opportunity to learn about different cultures, social perspectives, and expand my creative thinking. These courses challenged me in different ways that were not already present in my current engineering coursework and have expanded my ability to formulate engineering solutions with a more creative and socially conscious mindset.

In WISE 2010, I was able to explore the barriers faced by underrepresented groups in STEM fields. We focused on women in leadership positions, in particular. Through weekly discussions and presentations, we explored the structural barriers that limit who advance into leadership roles and how these barriers influence not only individual careers but also the quality and direction of innovation.

Learning about these dynamics helped me recognize that diverse leadership is not just a social goal, but directly affects how effectively engineering teams identify risks, ethical concerns, user needs, and underlying issues that other groups may overlook. I have carried this awareness with me into new work environments. I now pay closer attention to whose perspectives are included in the decision-making of engineering problems and how that shapes the solutions we create.

In my music courses, MUSIC 2460 and MUSIC 3460, I have been able to explore a more creative engineering process through experience in new digital tools, structured creative assignments, and interface design. In MUSIC 2460, I learned the fundamentals in audio waveform editing, mixing, and MIDI data structures. I was required to break down complex ideas into smaller components and iterate through smaller components and skills before tackling a larger creative project combining the skills, like the engineering design cycle.

In MUSIC 3460, I was able to expand my foundational skills by introducing computer music programming and interface design. While I have a background in a multitude of programming languages, creating musical applications using a visual language pushed me into new technical constraints. Both courses have enforced the idea of creative engineering and how innovative solutions often emerge from the ability to explore patterns, test ideas, and refine designs through experimentation.

While studying for my engineering degree, I had the opportunity to study abroad in Ireland. I not only could satisfy and further my learning in my core computer technology-based coursework, but I could also explore Irish cultures and customs. While abroad I took an

Irish Folklore course where I got to learn about Irish stories and traditions and connect how folklore connects to present-day perceptions.

In class we often discussed the Irish belief of preserving “thin places,” places that are viewed as spiritually significant or historically meaningful: ringforts, ancient pathways, or fairy tress for example. I remember we talked about a real case where a project was planned in County Clare but then redesigned because it would have required removing a hawthorn tree or “fairy tree” from the local community. Even though the tree had no technical impact on the road to be built, engineers rerouted to avoid disturbing the tree. Having a global perspective has been essential in my engineering career, as I am better equipped to produce solutions that are adaptable to diverse communities and different cultural contexts or societal expectations than my own.

This awareness also connects directly with the major societal challenge and the rise of ethical concerns as AI-driven tools, especially in the security space, are being introduced. As AI becomes more embedded in threat detection, identity verification, and risk scoring, engineers need to be more diligent about ensuring these systems are transparent, fair, and accessible to diverse users. Through my general education experiences, especially learning about how cultural beliefs shape acceptance, I reinforce the importance of designing cyber security solutions that people can trust.

Even though I have a couple more credits of general education left to fulfill, my current experiences have already broadened my perspective as a cyber security engineer. These courses have strengthened my ability to approach problems with creativity, cultural awareness, and ethical responsibility.